代数空间的弱 CHOW 覆盖

Counting scope. The primary problem is Tag 089J, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H3: 从有限型分离代数空间构造由 H-拟射影概形给出的真满覆盖，不能仅引用概形的 Chow 引理。证明以去掉对角的多重纤维积构造闭包，利用估值提升证明真性，并对几何纤维点数归纳覆盖剩余闭集。核心覆盖对象与归纳装配由本证明完成，故保留为 H3 候选，待独立确认。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, spaces-cohomology.tex, lines 3280–3396. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Definition 0.1. Let $f : X \to S$ be a morphism of schemes.

(1) We say f is *quasi-projective* if f is of finite type and there exists an f-relatively ample invertible $\mathcal{O}_X$-module.
(2) We say f is *H-quasi-projective* if there exists a quasi-compact immersion $X \to \mathbf{P}^n_S$ over S for some n.[1]
(3) We say f is *locally quasi-projective* if there exists an open covering $S = \bigcup V_j$ such that each $f^{-1}(V_j) \to V_j$ is quasi-projective.

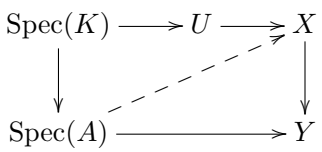
1. Projective morphisms

We will use the definition of a projective morphism from Bibliography: EGA. The version of the definition with the "H" is the one from Bibliography: H. The resulting definitions are different. Both are useful.

Lemma 1.1. *Let S be a scheme. Let $f : X \to Y$ and $h : U \to X$ be morphisms of algebraic spaces over S. If*

(1) *f and h are quasi-compact,*
(2) *$|h|(|U|)$ is dense in $|X|$, and*

given any commutative solid diagram



where A is a valuation ring with field of fractions K

(3) *there exists at most one dotted arrow making the diagram commute, and*

[1] This is not exactly the same as the definition in Hartshorne. Namely, the definition in Hartshorne (8th corrected printing, 1997) is that f should be the composition of an open immersion followed by a H-projective morphism (see Definition 1.4), which does not imply f is quasi-compact. See Lemma lemma H quasi projective open H projective (Tag 01WA) for the implication in the other direction.

(4) *there exists an extension K'/K of fields, a valuation ring $A' \subset K'$ dominating A and a morphism $\mathrm{Spec}(A') \to X$ such that the following diagram commutes*

$$\begin{array}{ccccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & U & \longrightarrow & X \\ \downarrow & & & & & & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

then f is universally closed. If moreover

(5) *f is quasi-separated*

then f is separated and universally closed.

Lemma 1.2. *Let S be a scheme. Let X be an algebraic space over S. Consider the following conditions on X:*

(α) *For every $x \in |X|$, the equivalent conditions of Lemma lemma U finite above x (Tag 03JS) hold.*

(β) *For every $x \in |X|$, the equivalent conditions of Lemma lemma R finite above x (Tag 03JU) hold.*

(γ) *For every $x \in |X|$, the equivalent conditions of Lemma lemma UR finite above x (Tag 03JV) hold.*

(δ) *The equivalent conditions of Lemma lemma U universally bounded (Tag 03JT) hold.*

(ϵ) *The equivalent conditions of Lemma lemma characterize very reasonable (Tag 03IH) hold.*

(ζ) *The space X is Zariski locally quasi-separated.*

(η) *The space X is quasi-separated*

(θ) *The space X is representable, i.e., X is a scheme.*

(ι) *The space X is a quasi-separated scheme.*

We have

$$\begin{array}{ccccccccc} & & (\theta) & & & & & & \\ & \nearrow & & \searrow & & & & & \\ (\iota) & & & & (\zeta) & \Longrightarrow & (\epsilon) & \Longrightarrow & (\delta) \Longrightarrow (\gamma) \Longleftrightarrow (\alpha) + (\beta) \\ & \searrow & & \nearrow & & & & & \\ & & (\eta) & & & & & & \end{array}$$

Lemma 1.3. *Let S be a scheme. Let $W \to X$ be a morphism of a scheme W to an algebraic space X which is flat, locally of finite presentation, separated, locally quasi-finite with universally bounded fibres. There exist reduced closed subspaces*

$$\emptyset = Z_{-1} \subset Z_0 \subset Z_1 \subset Z_2 \subset \ldots \subset Z_n = X$$

such that with $X_r = Z_r \setminus Z_{r-1}$ the stratification $X = \coprod_{r=0,\ldots,n} X_r$ is characterized by the following universal property: Given $g : T \to X$ the projection $W \times_X T \to T$ is finite locally free of degree r if and only if $g(|T|) \subset |X_r|$.

Definition 1.4. Let $f : X \to S$ be a morphism of schemes.

(1) We say f is *projective* if X is isomorphic as an S-scheme to a closed subscheme of a projective bundle $\mathbf{P}(\mathcal{E})$ for some quasi-coherent, finite type $\mathcal{O}_S$-module $\mathcal{E}$.

(2) We say f is *H-projective* if there exists an integer n and a closed immersion $X \to \mathbf{P}^n_S$ over S.
(3) We say f is *locally projective* if there exists an open covering $S = \bigcup U_i$ such that each $f^{-1}(U_i) \to U_i$ is projective.

Primary problem: Tag 089J

Lemma 1.5. *Let A be a ring. Let X be an algebraic space over $\mathrm{Spec}(A)$ whose structure morphism $X \to \mathrm{Spec}(A)$ is separated of finite type. Then there exists a proper surjective morphism $X' \to X$ where X' is a scheme which is H-quasi-projective over $\mathrm{Spec}(A)$.*

证明. Let W be an affine scheme and let $f : W \to X$ be a surjective étale morphism. There exists an integer d such that all geometric fibres of f have $\leq d$ points (because X is a separated algebraic hence reasonable, see Decent Spaces, Lemma 1.2). Picking d minimal we get a nonempty open $U \subset X$ such that $f^{-1}(U) \to U$ is finite étale of degree d, see Decent Spaces, Lemma 1.3. Let

$$V \subset W \times_X W \times_X \ldots \times_X W$$

(d factors in the fibre product) be the complement of all the diagonals. Because $W \to X$ is separated the diagonal $W \to W \times_X W$ is a closed immersion. Since $W \to X$ is étale the diagonal $W \to W \times_X W$ is an open immersion, see Morphisms of Spaces, Lemmas lemma etale unramified (Tag 06CR) and lemma diagonal unramified morphism (Tag 05W1). Hence the diagonals are open and closed subschemes of the quasi-compact scheme $W \times_X \ldots \times_X W$. In particular we conclude V is a quasi-compact scheme. Choose an open immersion $W \subset Y$ with Y H-projective over A (this is possible as W is affine and of finite type over A; for example we can use Morphisms, Lemmas lemma quasi affine finite type over S (Tag 04II) and lemma H quasi projective open H projective (Tag 01WA)). Let

$$Z \subset Y \times_A Y \times_A \ldots \times_A Y$$

be the scheme theoretic image of the composition $V \to W \times_X \ldots \times_X W \to Y \times_A \ldots \times_A Y$. Observe that this morphism is quasi-compact since V is quasi-compact and $Y \times_A \ldots \times_A Y$ is separated. Note that $V \to Z$ is an open immersion as $V \to Y \times_A \ldots \times_A Y$ is an immersion, see Morphisms, Lemma lemma quasi compact immersion (Tag 01RG). The projection morphisms give d morphisms $g_i : Z \to Y$. These morphisms g_i are projective as Y is projective over A, see material in Morphisms, Section 1. We set

$$X' = \bigcup g_i^{-1}(W) \subset Z$$

There is a morphism $X' \to X$ whose restriction to $g_i^{-1}(W)$ is the composition $g_i^{-1}(W) \to W \to X$. Namely, these morphisms agree over V hence agree over $g_i^{-1}(W) \cap g_j^{-1}(W)$ by Morphisms of Spaces, Lemma lemma equality of morphisms (Tag 084N). Claim: the morphism $X' \to X$ is proper.

If the claim holds, then the lemma follows by induction on d. Namely, by construction X' is H-quasi-projective over $\mathrm{Spec}(A)$. The image of $X' \to X$ contains the open U as V surjects onto U. Denote T the reduced induced algebraic space structure on $X \setminus U$. Then $T \times_X W$ is a closed subscheme of W, hence affine. Moreover, the morphism $T \times_X W \to T$ is étale and every geometric fibre has $< d$ points. By induction hypothesis there exists a proper surjective morphism $T' \to T$ where T' is a scheme H-quasi-projective over $\mathrm{Spec}(A)$. Since T is a closed subspace of X we see that $T' \to X$ is a proper morphism. Thus the lemma follows by taking the proper surjective morphism $X' \amalg T' \to X$.

Proof of the claim. By construction the morphism $X' \to X$ is separated and of finite type. We will check conditions (1) – (4) of Morphisms of Spaces, Lemma 1.1 for the morphisms $V \to X'$ and $X' \to X$. Conditions (1) and (2) we have seen above. Condition (3) holds as $X' \to X$ is separated (as a morphism whose source is a separated algebraic space). Thus it suffices to check liftability to X' for diagrams

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \longrightarrow & X \end{array}$$

where R is a valuation ring with fraction field K. Note that the top horizontal map is given by d pairwise distinct K-valued points $w_1, \ldots, w_d$ of W. In fact, this is a complete set of inverse images of the point $x \in X(K)$ coming from the diagram. Since $W \to X$ is surjective, we can, after possibly replacing R by an extension of valuation rings, lift the morphism $\mathrm{Spec}(R) \to X$ to a morphism $w : \mathrm{Spec}(R) \to W$, see Morphisms of Spaces, Lemma lemma lift valuation ring through flat morphism (Tag 089F). Since $w_1, \ldots, w_d$ is a complete collection of inverse images of x we see that $w|_{\mathrm{Spec}(K)}$ is equal to one of them, say w_i. Thus we see that we get a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & Z \\ \downarrow & & \downarrow{\scriptstyle g_i} \\ \mathrm{Spec}(R) & \xrightarrow{w} & Y \end{array}$$

By the valuative criterion of properness for the projective morphism g_i we can lift w to $z : \mathrm{Spec}(R) \to Z$, see Morphisms, Lemma lemma locally projective proper (Tag 01WC) and Schemes, Proposition proposition characterize universally closed (Tag 01KF). The image of z is in $g_i^{-1}(W) \subset X'$ and the proof is complete. $\square$